

14th day of class - 10/9

Fun fact: More about the j -invariant.

Given two lattices L_1 and L_2 in $\mathbb{C}$, the corresponding elliptic curves are isomorphic over $\mathbb{C}$ if L_1 is a scaling of L_2 . (There is a constant $c \in \mathbb{C}$ so that $L_2 = \{cz : z \in L_1\}$.) This follows from staring at the formulas for the g_2 and g_3 functions.

Given a lattice $L_1 = \{a\omega_1 + b\omega_2 : a, b \in \mathbb{Z}\}$ we can scale it so one of the basis numbers is equal to 1. For example, we could scale by $1/\omega_1$ and get $\{a + b\omega_2/\omega_1 : a, b \in \mathbb{Z}\}$.

For a complex number z (with nonzero imaginary part) define $j(z)$ to be the j -invariant of the elliptic curve with lattice $\{1 + bz : a, b \in \mathbb{Z}\}$. Since $j(z) = j(z+1)$, we can represent $j(z)$ as a series in $q = e^{2\pi iz}$ and

$$j(z) = q^{-1} + 744 + 196884q + 21493760q^2 + \dots$$

(Every continuous periodic function can be represented as a Fourier series.)

Last time: Most of the proof of the Lutz-Nagell theorem.

Today: Finish the proof of the Lutz-Nagell theorem. Talk about heights?

Reminder of definitions: Let $C : y^2 = x^3 + ax^2 + bx + c$.

Define

$$C(p^\nu) = \{(x, y) \in C(\mathbb{Q}) : \text{ord}_p(x) \leq -2\nu \text{ and } \text{ord}_p(y) \leq -3\nu\} \cup \{(0 : 1 : 0)\}.$$

Claim 1: Each $C(p^\nu)$ is a subgroup of $C(\mathbb{Q})$!

Claim 2: If $P \in C(p^\nu)$ but P is not in $C(p^{\nu+1})$, then $mP \in C(p^{\nu+1})$ if and only if $p|m$. Moreover, $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$.

For $P \in C(\mathbb{Q})$, write $P = (x, y)$ and define $t(P) = x/y$. We have $P \in C(p^\nu)$ if and only if $t(P) \equiv 0 \pmod{p^\nu}$.

Theorem. If $P_1, P_2 \in C(p^\nu)$, then

$$t(P_1 + P_2) \equiv t(P_1) + t(P_2) \pmod{p^{3\nu}}.$$

Proof of Claim 1. The congruence $t(P_1 + P_2) \equiv t(P_1) + t(P_2) \pmod{p^{3\nu}}$ only assumes that P_1 and P_2 are in $C(p^\nu)$. If P_1 and P_2 are in $C(p^\nu)$, then $t(P_1) \equiv t(P_2) \equiv 0 \pmod{p^\nu}$. Thus

$$t(P_1 + P_2) \equiv 0 \pmod{p^\nu}$$

and this implies that $P_1 + P_2 \in C(p^\nu)$. Also, if $P \in C(p^\nu)$ and $P = (x, y)$, then $-P = (x, -y) \in C(p^\nu)$ also. $\square$

To prove Claim 2, we need to make the t -map into a homomorphism. We just showed the domain is a subgroup of $C(\mathbb{Q})$.

Definition. Let $R = \{r \in \mathbb{Q} : \text{ord}_p(r) \geq 0\}$. This is the set of rational numbers whose denominator is not a multiple of p .

Note that R is closed under addition, because if

$$r_1, r_2 \in R$$

then $r_1 = \frac{a}{b}$ and $r_2 = \frac{c}{d}$ with $\gcd(a, b) = \gcd(c, d) = \gcd(b, p) = \gcd(d, p) = 1$. We get

$$r_1 + r_2 = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

So the denominator is a divisor of bd , but b and d are both coprime to p . (The set R is also closed under multiplication.)

We don't have $t(P_1 + P_2) = t(P_1) + t(P_2)$ - this equality is only true modulo $p^{3\nu}$. Also, since P_1 and $P_2 \in C(p^\nu)$ we have $t(P_1) \equiv t(P_2) \equiv 0 \pmod{p^\nu}$.

So, what we do is we define $p^k R = \{p^k r : r \in R\}$. This is a subgroup of $p^k R$. We can think of the map t as a homomorphism

$$t : C(p^\nu) \rightarrow p^\nu R / p^{3\nu} R.$$

The right hand side is a cyclic group of order $p^{2\nu}$.

Q: Is t surjective?

A: Not necessarily. There are examples where the map $t : C(p) \rightarrow pR/p^3R$ is just the zero map.

Q: What's the kernel of t ?

A: We have that P is in the kernel of $t : C(p^\nu) \rightarrow p^\nu R / p^{3\nu} R$ if and only if $t(P) \equiv 0 \pmod{p^{3\nu}}$. If we write $P = (a/p^{2k}, b/p^{3k})$ where $\text{ord}_p(a) = \text{ord}_p(b) = 0$, then $t(P) = \frac{a/p^{2k}}{b/p^{3k}} = p^k(a/b)$ so $\text{ord}_p(t(P)) = k$. Thus, $t(P) \equiv 0 \pmod{p^{3\nu}}$ if and only if $\text{ord}_p(t(P)) \geq 3\nu$, that is, if $P \in C(p^{3\nu})$.

Thus, $C(p^{3\nu})$ is the kernel of t .

So the first isomorphism theorem means that we have an injective homomorphism

$$C(p^\nu)/C(p^{3\nu}) \rightarrow p^\nu R / p^{3\nu} R.$$

Claim 2: If $P \in C(p^\nu)$ but P is not in $C(p^{\nu+1})$, then $mP \in C(p^{\nu+1})$ if and only if $p|m$. Moreover, $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$.

Proof of Claim 2. If $P \in C(p^\nu)$ and $P \notin C(p^{\nu+1})$, then $\text{ord}_p(t(P)) = \nu$.

We have that $mP \in C(p^{\nu+1})$ if and only if $t(mP) \equiv 0 \pmod{p^{\nu+1}}$. But $t(mP) \equiv mt(P) \pmod{p^{3\nu}}$ and so $t(mP) \equiv mt(P) \pmod{p^{\nu+1}}$. This proves that $mP \in C(p^{\nu+1})$ if and only if $p|m$.

Now, we in fact have $t(mP) \equiv mt(P) \pmod{p^{\nu+2}}$ (since $3\nu \geq \nu + 2$ if $\nu \geq 1$). So if $t(P) \equiv 0 \pmod{p^\nu}$ but $t(P) \not\equiv 0 \pmod{p^{\nu+1}}$, then

$$t(pP) \equiv pt(P) \pmod{p^{\nu+2}}$$

must be $\equiv 0 \pmod{p^{\nu+1}}$, but $pt(P) \not\equiv 0 \pmod{p^{\nu+2}}$. Thus, $\text{ord}_p(t(pP)) = \nu + 1$ and so $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$. $\square$

Example (final exam from 2015): Let $C : y^2 = x^3 - 16x + 16$ and let $P = (0, 4)$.

(a) What is the smallest n so that $nP \in C(11)$?

We use Magma. The points kP for $1 \leq k \leq 6$ are integral. We keep checking and eventually find that $17P \in C(11)$ with $kP \notin C(11)$ for $1 \leq k \leq 16$. We can also compute that $t(17P) \equiv 110 \pmod{121}$.

(b) What is the smallest n so that $nP \in C(11^5)$?

We don't need Magma anymore.

We have that $17P \in C(11)$ but that $kP \notin C(11)$ for $1 \leq k \leq 16$. Also the x and y -coordinates of $17P$ are

$$11^2 \cdot 139^2 \text{ and } 11^3 \cdot 139^3,$$

respectively. Thus, $17P \notin C(11^2)$.

If $nP \in C(11)$, then if we write $n = 17q + r$ for $0 \leq r \leq 16$, then we have $nP = q(17P) + rP$. We know that $17P \in C(11)$ and thus $rP = nP - q(17P) \in C(11)$ since $C(11)$ is a group. This implies that $r = 0$. It follows that $C(11) = \{17qP : q \in \mathbb{Z}\}$.

Claim: For any positive integer k , we have that $nP \in C(11^k)$ if and only if $17 \cdot 11^{k-1} \mid n$.

We have proven the base case of $k = 1$.

Assume that we the k th case is known. Then $17 \cdot 11^{k-1}P \in C(11^k)$. Claim 2 implies that $17 \cdot 11^kP \in C(11^{k+1})$ and since $C(11^{k+1})$ is a subgroup of $C(\mathbb{Q})$, if $17 \cdot 11^k \mid n$, then $nP \in C(11^{k+1})$.

Conversely, if $nP \in C(11^{k+1})$, then $nP \in C(11^k)$ and so n is a multiple of $17 \cdot 11^{k-1}$. The inductive hypothesis implies that $17 \cdot 11^{k-1}rP \notin C(11^{k+1})$ unless $11 \mid r$ and from this it follows that $nP \in C(11^{k+1})$ if and only if $17 \cdot 11^k \mid n$.

QED Claim

It follows that the smallest n so that $nP \in C(11^5)$ is $n = 17 \cdot 11^4$.

The Mordell-Weil theorem

Theorem. Suppose that $C : y^2 = x^3 + ax^2 + bx + c$ is an elliptic curve and $a, b, c \in \mathbb{Z}$. Then $C(\mathbb{Q})$ is finitely generated.

The Mordell-Weil theorem is based on the idea of “descent.” This method allows one to find a finite set of points $\{Q_1, Q_2, \dots, Q_n\}$ so that if $P \in C(\mathbb{Q})$ is any point, then

1. there is some Q_{i_1} so that $P - Q_{i_1} = 2P_1$ for some $P_1 \in C(\mathbb{Q})$.
2. the point P_1 will be much simpler than P .
3. We can do the same thing to P_1 and write $P_1 - Q_{i_2} = 2P_2$ for some $P_2 \in C(\mathbb{Q})$.
4. the point P_2 will be much simpler than P_1 .

continue.

Q: What do we mean by simple?

A: If $x \in \mathbb{Q}$ and we write $x = \frac{m}{n}$ with $\gcd(m, n) = 1$, we define $H(x) = \max\{|m|, |n|\}$.

If $P = (x, y) \in C(\mathbb{Q})$, we define $H(P) = H(x)$ (and we define $H((0 : 1 : 0)) = 1$). Finally, we define $h(P) = \ln H(P)$.

Key property: If $M > 0$, then there are only finitely many $x \in \mathbb{Q}$ so that $H(x) < M$.

The four lemmas

Lemma. *For any real number M , the set $\{P \in C(\mathbb{Q}) : h(P) \leq M\}$ is finite.*

Proof. We know that there are only finitely many $x \in \mathbb{Q}$ so that $H(x) \leq e^M$. For each such x , there are at most 2 values of y so that $(x, y) \in C(\mathbb{Q})$. Thus, this set is finite. $\square$

Lemma. *Let P_0 be a fixed point on C . There is a constant κ_0 , depending on P_0 , a , b and c so that*

$$h(P + P_0) \leq 2h(P) + \kappa_0 \quad \forall P \in C(\mathbb{Q}).$$

Lemma. *There is a constant κ , depending on a , b and c so that*

$$h(2P) \geq 4h(P) - \kappa \quad \forall P \in C(\mathbb{Q}).$$

This means that $H(2P) \approx H(P)^4$.

Definition. *If G is a group and H is a subgroup of G , then index of H in G $[G : H]$ is the number of distinct left cosets of H .*

If G is finite, every left coset of H has size equal to $|H|$ and every element of G is in exactly one left coset. Then,

$$[G : H] = \frac{|G|}{|H|}.$$

However, we will apply this idea with infinite groups also.

Let $2C(\mathbb{Q}) = \{P + P : P \in C(\mathbb{Q})\}$. This is a subgroup of $C(\mathbb{Q})$.

Lemma. *The index $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ is finite.*

These four lemmas will prove the Mordell-Weil theorem, and they are listed in increasing order of difficulty. The first one we have already proven. Lemmas 2 and 3 will require a detailed analysis of the group law. Lemma 4 is the most difficult, and we'll only prove it in the special case of curves of the form $y^2 = x^3 + ax^2 + bx$, that is, curves with a rational point of order 2.